GAJA SAVIYA - An Open-Data, Software-Only Early Warning and Safe-Routing Framework for Human-Elephant Conflict in Rural Sri Lanka

INDIVIDUAL THESIS

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Individual component: Risk-Aware Safe Routing for Human–Elephant Conflict Mitigation

Group ID: R26-IT-136

Report submitted in partial fulfilment of the requirements for the

B.Sc (Hons) Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026

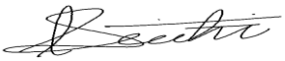
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ABSTRACT

Human–elephant conflict (HEC) in rural Sri Lanka causes serious harm to both people and elephants, particularly where habitats overlap with farms, villages, and transport routes. Existing mitigation approaches rely heavily on physical sensors, official GPS collar data, or static hotspot maps — resources that are often unavailable or expensive for university-level software prototypes. This individual thesis addresses Component 3 (Risk-Aware Safe Routing) of the group Gaja-Saviya framework (Ali stack implementation), an open-data, software-only early-warning system developed for Anuradhapura District.

The component designs, implements, and evaluates a road-network routing mechanism that incorporates spatial HEC risk information from Component 1 (seasonal XGBoost risk surface, 500 m grid, EPSG:5234) and tracker uncertainty from Component 2 (Extended Kalman Filter estimates between sparse simulated GPS updates). Road data are drawn from OpenStreetMap; risk is mapped to road segments using length-weighted intersection with the risk grid; and an edge-cost function combining distance, mean risk, and mean uncertainty selects safer alternatives using Dijkstra's algorithm with non-negative weights.

The evaluation compares a conventional shortest-distance route with a risk-aware route across selected origin–destination pairs within the study area. Metrics include total distance, additional travel distance, high-risk route exposure (proportion of route passing through cells above a predefined risk threshold), and sensitivity to the risk weight (α) and uncertainty weight (β). Because this is a computational prototype using simulated elephant movement (transfer statistics from published Kruger dataset via Movebank, applied to Sri Lankan landscape), findings demonstrate methodological feasibility rather than real-world operational safety.

At the time of writing, the component is at the design and planning stage; implementation, experimental execution, and full validation are in progress. The thesis contributes a transparent, reproducible routing model that connects spatial risk estimation and movement uncertainty to actionable route recommendations, and it documents all assumptions, data sources, parameter settings, and limitations required for independent review.

Keywords: human–elephant conflict; risk-aware routing; species distribution modelling; Extended Kalman Filter; OpenStreetMap; safe routing; Anuradhapura; software prototype.

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my research supervisor for the guidance, feedback, and support provided throughout this research. I also thank the lecturers and the Sri Lanka Institute of Information Technology for the knowledge and academic support that contribute to my research and learning.

I am grateful to my project team members for their cooperation in developing the broader research framework. Finally, I would like to thank my family and everyone who has encouraged and supported me during my academic journey.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
Dijkstra	Dijkstra's shortest-path algorithm
GIS	Geographic Information System
HEC	Human–Elephant Conflict
ML	Machine Learning
OSM	OpenStreetMap
XGBoost	Extreme Gradient Boosting
WGS84	World Geodetic System 1984

1 INTRODUCTION

1.1 Overall project context

Human–elephant conflict is a continuing concern in Sri Lanka, particularly in rural and Dry Zone settings where elephant habitat can overlap with farms, villages, and other human activities. The research proposal describes a practical problem: people may receive information about an elephant only after it has reached a village or farm. Existing approaches may depend on equipment that is difficult for a student team to install, or on official GPS-collar and conflict-incident data that are not generally available for open student use.

The overall research project proposes a software-only early-warning and safe-routing framework for Anuradhapura District. It connects four components: a seasonal HEC proxy-risk map; a tracker that estimates simulated elephant movement between infrequent GPS-like updates; a road-corridor risk component that highlights nearby roads and provides advice; and a Path Guard service that compares a shortest route with a safer route. The system is designed for a repeatable demonstration using public data and explicitly labelled simulations.

1.2 Individual component context

This thesis focuses on Component 3, titled in the proposal as “Risk on road corridors, nearby roads and advice.” The component converts public OpenStreetMap roads into a connected road network and divides the network into road sections for machine learning. A separate XGBoost model predicts segment_risk, described as a proxy probability of conflict exposure for each section. The model uses information from Component 1, road or highway type, hour of day, and distance from or overlap with the uncertainty circle from Component 2.

The main local-roads demonstration highlights roads near the estimated elephant location. Roads inside or close to the uncertainty circle, or roads with extremely high land risk, are marked red; roads farther away are marked green unless land risk is extreme. The service also provides structured advice in clear English, including a

headline, road names to avoid, and nearby places. An optional A-to-B routing method uses a risk-weighted Dijkstra approach. Component 4, Path Guard, remains a separate service that synchronises the road network and compares shortest and safer A-to-B routes.

1.3 Problem statement

A map showing broad HEC hotspots may not tell a person which nearby road sections should be avoided when an elephant's location is uncertain. The research proposal identifies a missing practical link between geographic risk information and understandable warnings about particular roads. Component 3 addresses this gap by turning risk and estimated-location information into road-section scores, map colours, and plain-English advice.

The individual problem addressed by this thesis is how to use public road data, a modelled HEC proxy-risk surface, and simulated location estimates to classify nearby road sections and support risk-aware route comparison without relying on official Sri Lankan elephant collar records or confirmed incident labels.

1.4 Research gap

The proposal identifies several limitations in common approaches. Field sensors and camera traps may involve cost, maintenance, access, and ethical considerations. Official Department of Wildlife Conservation collar and incident records are not generally openly downloadable for students. Static hotspot maps may not be connected to changing animal-location estimates, uncertainty, warnings about particular roads, or route suggestions. A routing method that only applies Dijkstra to grid-cell risk may also fail to learn risk for individual road corridors or to give advice that names nearby roads.

The gap addressed by Component 3 is a practical road-level risk service that uses public data and clearly identified proxy labels to score road sections, highlight nearby roads in relation to an estimated elephant location, and provide understandable

advice. This is a computational prototype, not a claim that the service has been validated as an operational wildlife-warning system.

1.5 Aim and objectives

Aim: To design, implement, and evaluate a software-based road-corridor risk component that uses public road data, proxy HEC risk information, and estimated elephant-location uncertainty to highlight potentially threatened roads and support safer route selection in Anuradhapura District.

The individual objectives are:

1. Prepare a connected road network from public OpenStreetMap data for the Anuradhapura demonstration area.
2. Develop or use a separate XGBoost road-section model to estimate proxy conflict-exposure risk for road sections.
3. Use land and risk information from Component 1, road or highway type, time of day, and the Component 2 uncertainty area as model inputs, as specified in the proposal.
4. Provide a local-roads service that highlights nearby road sections and generates clear-English advice about roads to avoid.
5. Support optional comparison of a distance-based shortest route with a risk-weighted Dijkstra route.
6. Evaluate road warnings and route comparisons using numerical tests and simulated scenarios, without representing proxy labels as confirmed elephant incidents.

1.6 Research questions or hypotheses

This individual research is guided by the following research question:

RQ3: Can a trained road-corridor risk model, together with distance from the tracker's uncertainty area, produce useful road colours and warnings? Can Path

Guard reduce travel through highly threatened road sections, and how much extra distance does the safer route require?

For this individual thesis, the question is addressed through the road-corridor model and local-roads API, while the optional routing comparison is evaluated as a supporting capability. No claim of reduced real-world conflict is made.

1.7 Scope limitations and assumptions

The work is limited to Anuradhapura District as the fixed demonstration location. It uses public datasets and project-generated simulated paths. OpenStreetMap provides road and related vector-map information. Component 1 supplies the proxy-risk context, and Component 2 supplies simulated elephant-location estimates and an uncertainty circle. Road-model labels are proxies derived from high-risk corridors and indicators such as farmland and settlement edges; they are not official HEC incident records.

The study does not involve capturing, collaring, approaching, or otherwise interacting with wild elephants. It does not install sensors on private land or inside forests, use restricted Department of Wildlife Conservation collar data, or claim to replace official wildlife-warning systems. It also does not establish real-time location accuracy for any particular wild elephant in Sri Lanka.

1.8 Individual contribution

The individual contribution covered by this thesis is Component 3: the road-corridor risk model, local-roads API, road warning presentation and advice, and optional risk-weighted routing comparison. The wider framework's seasonal danger map, sparse tracker, Path Guard service, and Next.js interface are treated as connected components rather than as the primary subject of this individual thesis.

Before final submission, the author should update this section with a precise, evidence-based record of the code, data preparation, experiments, and integration tasks personally completed. The proposal alone does not specify the division of individual implementation tasks in enough detail to make additional claims.

1.9 Thesis structure

Chapter 1 introduces the project context, individual problem, research gap, aim, objectives, research question, scope, and contribution.

Chapter 2 reviews the domain concepts and the approaches identified in the proposal.

Chapter 3 describes the research design, data, procedure, evaluation metrics, responsible use, and reproducibility.

Chapter 4 explains the requirements and design of Component 3.

Chapter 5 documents implementation, integration, testing, and operation.

Chapter 6 presents research and technical results, with measured results to be inserted only after tests are run.

Chapter 7 discusses the findings and limitations.

Chapter 8 concludes and identifies future work.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This chapter is a proposal-grounded review of the concepts and limitations explicitly described in the supplied research proposal. The proposal names datasets, modelling methods, and broad categories of existing approaches, but it does not provide full bibliographic details for individual research papers. Accordingly, this chapter does not invent author names, publication titles, years, or performance comparisons. Before formal submission, the author must add a systematic literature search and complete citations for the underlying sources required by the department.

2.2 Domain and theoretical foundations

2.2.1 Human–elephant conflict as a spatial problem

The proposal frames HEC risk as a consequence of overlap between suitable elephant habitat and human exposure. Forest presence alone is not sufficient to indicate conflict. Farms, built-up areas, population, settlement edges, and seasonal factors are relevant to the project's proxy-risk representation. Component 1 provides a seasonal spatial risk map that Component 3 can use to inform road-section risk.

2.2.2 Road-corridor risk

A road network consists of connected road segments. Assigning a score to each segment makes it possible to distinguish roads that are more or less exposed to the proxy risk surface. In this project, segment_risk is a model estimate for a road section, not an observed count of elephant incidents. This distinction matters because the model's labels are generated from risk corridors and geographic indicators rather than verified incident records.

2.2.3 XGBoost for road-section classification

The proposal specifies a separate XGBoost model for estimating risk on road sections. Its input features include land and risk values from Component 1, road/highway type, hour of day, and distance from or overlap with the uncertainty circle from

Component 2. The proposal does not provide the complete feature schema, hyperparameters, training split, calibration procedure, or Component 3 evaluation scores. Those details must be documented from the actual model pipeline and experiment logs.

2.2.4 Shortest-path and risk-weighted routing

A conventional shortest-path method seeks a route with minimum distance. The proposal's optional Component 3 A-to-B method uses a risk-weighted Dijkstra approach with edge cost defined as length × (1 + λ × segment_risk). The parameter λ controls how strongly the route cost penalises risk relative to distance. Component 4's Path Guard separately compares a shortest route with a safer route and integrates threat information into the user-facing flow.

2.2.5 Location uncertainty

Component 2 uses an Extended Kalman Filter to estimate elephant location between infrequent simulated GPS-like updates and represents uncertainty using a circle. Component 3 uses distance from or overlap with this uncertainty area as part of the road-section risk model. The road-warning logic also marks roads red when they are inside or close to the uncertainty circle, subject to the proposal's described land-risk rule.

2.3 Related research

The supplied proposal identifies movement statistics from a published African elephant GPS dataset (Kruger / Movebank) as the basis for Component 2's simulated movement, while explicitly stating that these statistics are not real Sri Lankan collar records. It also identifies GBIF/iNaturalist elephant-presence records, ESA WorldCover, OpenStreetMap, WorldPop, NASA POWER, Copernicus GLO-30 DEM, and protected-area information as sources used across the overall framework.

For Component 3, the most relevant related-work themes are road-level wildlife-risk estimation, geospatial feature engineering, uncertainty-aware risk representation, interpretable warning delivery, and risk-aware shortest-path methods. The proposal

gives the project rationale for these themes but does not provide enough detailed paper-level evidence to compare individual algorithms or claim superiority over prior studies.

2.4 Existing systems products and techniques

Approach identified in the proposal	**Limitation for this project**	**Relevance to Component 3**
IoT sensors and camera traps	Cost, possible damage or theft, maintenance, ethical questions, and limited field access.	Supports a software-only approach based on public data and simulation.
Official wildlife collar and incident databases	Usually not openly available for students to download.	Requires clearly labelled proxy labels and simulated locations.
Static GIS hotspot maps	May not connect changing animal locations and uncertainty to road-specific warnings or A-to-B routes.	Component 3 links spatial risk to individual road sections.
Dijkstra using only grid-cell risk	May not learn risk for specific road corridors or name nearby roads in advice.	The proposal uses a separate road-section model and a local-roads service.
Distance-only shortest route	Minimises distance but does not explicitly penalise road-section risk.	Provides the baseline for optional risk-weighted route comparison.

Table-01

2.5 Gap synthesis and conceptual framework

The proposal's central gap is the missing practical connection between spatial HEC risk, an estimated and uncertain elephant location, warnings about particular roads, and route suggestions. Component 3 addresses the road-level part of this connection. It receives risk context from Component 1 and location/uncertainty information from Component 2; it produces road-section scores, local road colours, and plain-English advice. Its optional risk-weighted route calculation can also provide a comparison baseline for the wider Path Guard service.

The conceptual flow for Component 3 is: public OpenStreetMap road network + Component 1 land/risk values + road attributes and time + Component 2 estimated location and uncertainty → road-section features → XGBoost proxy-risk estimate → nearby-road colours and advice → optional risk-weighted Dijkstra comparison. This framework should be illustrated with a figure derived from the actual implementation.

A key methodological boundary is that model output represents estimated proxy exposure. It should not be described as a confirmed probability of an elephant encounter unless a validated label source and calibration process support that interpretation.

3 RESEARCH METHODOLOGY

3.1 Research design

This work follows an applied research and computational-prototype design. The overall project is software-only and uses public data together with controlled simulation. Component 3 is assessed by checking whether the road-section model and local-roads service produce the intended outputs and by comparing distance-only routing with optional risk-weighted routing under defined scenarios.

The evaluation should use the same road network and origin–destination pairs for baseline and risk-aware route comparisons. The study should record the chosen study area, data sources, preprocessing steps, model configuration, risk-weight setting, high-risk classification threshold, software environment, and number of completed scenarios. These records are necessary for the results to be reproducible.

3.2 Data participants or study materials

Purpose	**Source stated in proposal**	**Use in Component 3**	**Important qualification**
Road network and road attributes	OpenStreetMap (Overpass / Geofabrik)	Road geometries, network connections, road/highway type, and road names where available.	Record the actual extract date, coverage, and download method.
Land cover	ESA WorldCover	Land/risk features inherited from Component 1.	Record actual layers and preprocessing used.
Human exposure	WorldPop	Population-related exposure features through Component 1.	Not a direct record of HEC incidents.
Seasonal climate	NASA POWER	Monthly rainfall/temperature and seasonal context used in Component 1.	Record selected month and data period.

Terrain	Copernicus GLO-30 DEM	Potential terrain context if included in the implemented risk feature pipeline.	Include only if actually used by Component 3.
Elephant presence	GBIF / iNaturalist	Presence records inform Component 1 habitat suitability and risk context.	Presence records are not equivalent to verified conflict incidents.
Estimated elephant location	Project-generated simulated movement and Component 2 EKF	Estimated location and uncertainty circle for road proximity/overlap features.	Simulated data must be labelled as synthetic.
Road-section training labels	Project-generated proxy labels	Labels derived from high-risk corridors and farmland/settlement-edge indicators.	Not official or confirmed HEC incident records.

Table-02

3.3 Experimental or study procedure

1. Obtain and prepare the OpenStreetMap road extract for Anuradhapura District, retaining the attributes required by the road-section model and local-road advice.
2. Prepare or access the land and risk information produced by Component 1 and align it spatially with the road sections.
3. Obtain the estimated elephant location and uncertainty information from the simulated Component 2 tracker.
4. Construct the road-section feature set using the inputs specified in the proposal: land and risk values, road/highway type, hour of day, and distance from or overlap with the uncertainty circle.
5. Generate proxy labels using the documented rules based on high-risk corridors and farmland/settlement-edge indicators. Preserve the distinction between proxy labels and confirmed incidents.

6. Train or load the separate XGBoost road-section model and record the exact training, validation, and model configuration from the implementation.
7. Run the local-roads API for defined simulated elephant locations and times. Record road colours, risk scores, advice, road names, and nearby places where returned.
8. Where the optional A-to-B method is tested, calculate a distance-only shortest route and a risk-weighted Dijkstra route on the same network and origin–destination pair.
9. Calculate route distance, high-risk exposure, and risk-weighted route cost using a documented threshold and consistent definitions.
10. Repeat tests across predefined locations, times, and risk-weight settings. Record all completed scenarios and report the actual results without selecting only favourable examples.

3.4 Measures and evaluation metrics

Metric	**Definition or calculation**	**Purpose**
Road-section risk output	The model's predicted segment_risk for each road section.	Checks that the model returns a score for the road sections evaluated.
Road warning output	Red/green classification under the implemented proximity and land-risk rules.	Checks whether nearby roads are highlighted according to the stated rules.
Advice completeness	Whether the response includes the expected headline, road names to avoid, and nearby places when available.	Checks usefulness and completeness of structured advice.
Shortest route distance	Sum of physical road-segment lengths on the distance-only route.	Baseline distance for comparison.

Risk-aware route distance	Sum of physical road-segment lengths on the risk-weighted route.	Measures the travel-distance cost of risk avoidance.
Additional distance	D(risk-aware) − D(shortest).	Measures extra physical distance.
Percentage distance increase	((D(risk-aware) − D(shortest)) / D(shortest)) × 100.	Expresses extra distance relative to the baseline.
High-risk route share	(Distance inside high-risk sections / total route distance) × 100.	Compares route exposure to sections classified as high risk.
Risk-weighted route cost	Sum of edge costs, with edge cost defined in the proposal as length × (1 + λ × segment_risk).	Checks the route objective used by the optional routing method.

Table-03

3.5 Data analysis

The analysis should compare each shortest route and risk-weighted route within the same scenario. Route distance and high-risk route share should be reported side by side. If multiple scenarios are run, report per-scenario values and appropriate summary statistics, such as the mean, median, range, or distribution, where the sample size supports them. The analysis should also investigate how route selection changes with the risk weight λ.

For local-road warnings, results should be assessed against the component's documented decision rules. For example, a test should check whether a road inside or close to the uncertainty circle is highlighted as red unless the implemented rules specify an exception. Any accuracy or classification metrics for the XGBoost model must be based on a documented validation procedure and proxy labels; they must not be interpreted as real-world incident-prediction accuracy.

3.6 Ethics privacy security and responsible AI

The proposal's design avoids field interaction with wild elephants and does not use restricted Sri Lankan collar records. Simulated movement is clearly labelled as synthetic. Proxy labels must be described as proxies, not confirmed incidents. Outputs should be presented as decision-support information for a research demonstration, not as guaranteed safety advice or a replacement for official Department of Wildlife Conservation systems.

Public map data should be used in accordance with its applicable terms and attribution requirements. If the prototype exposes API endpoints or map data, deployment settings should be reviewed before any public hosting. The proposal does not provide a completed security assessment, user study, or privacy audit; these must not be claimed unless conducted.

3.7 Reproducibility and research artefacts

For reproducibility, preserve the road extract version/date, dataset sources, spatial reference systems, preprocessing steps, proxy-label generation rules, feature definitions, model file and configuration, λ values, risk thresholds, origin–destination pairs, simulated tracker inputs, API responses, and experiment logs. The proposal names OpenStreetMap, ESA WorldCover, WorldPop, NASA POWER, Copernicus GLO-30 DEM, GBIF/iNaturalist, and published Kruger/Movebank elephant GPS data as sources across the full framework. The final thesis should state exactly which of these sources were used by Component 3 and which were only used upstream by other components.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

4.1.1 Functional requirements

- The component shall accept or access an estimated elephant location and the associated uncertainty information from Component 2.
- The component shall evaluate road sections using the road network and the risk features available from Component 1 and Component 2.
- The component shall produce road-section risk outputs using the separate XGBoost road model described in the proposal.
- The local-roads service shall highlight nearby roads using the documented proximity and land-risk rules.
- The service shall provide structured advice in clear English, including a headline, roads to avoid, and nearby places when those details are available.
- The optional routing method shall support a comparison between a conventional shortest route and a route using risk-weighted edge costs.
- The system shall make clear that simulated locations and proxy risk labels are not confirmed real-world observations.

4.1.2 Non-functional requirements

- Repeatability: tests should be reproducible using recorded inputs, parameter settings, and data versions.
- Explainability: advice should use understandable wording and identify roads or places when the underlying data support it.
- Interoperability: the component should expose a service interface that can be called by other parts of the connected system.
- Robustness: missing data or invalid requests should be handled clearly by the service.
- Responsible use: the prototype should not imply guaranteed safety or real-time wildlife-tracking accuracy.

4.2 Component architecture within the complete solution

Component 3 is positioned between the risk-map and tracker components on one side and the Path Guard route service on the other. Component 1 provides a seasonal risk surface based on habitat suitability, human exposure, and a rainfall/season factor. Component 2 provides a simulated elephant journey, estimated location, and uncertainty. Component 3 combines this context with road-section features to predict proxy risk and return nearby-road colours and advice. Component 4 uses the road network and threat information to compare shortest and safer A-to-B routes through the main Next.js interface.

Input or connection	Information exchanged	Component 3 role
Component 1 — Danger map	Spatial land/risk values and risk context.	Provides geographic risk features for road sections.
Component 2 — Sparse tracker	Estimated elephant location and uncertainty circle.	Supports distance/overlap features and nearby-road warnings.
OpenStreetMap road data	Road geometry, connectivity, road class/name attributes where available.	Defines road sections to score and route over.
Component 3 model and local-roads API	segment_risk, road colours, structured advice.	Primary individual component output.
Component 4 — Path Guard	Road network and threat information for A-to-B comparison.	Consumes road/threat context; Path Guard is not the same service as Component 3.

Table-04

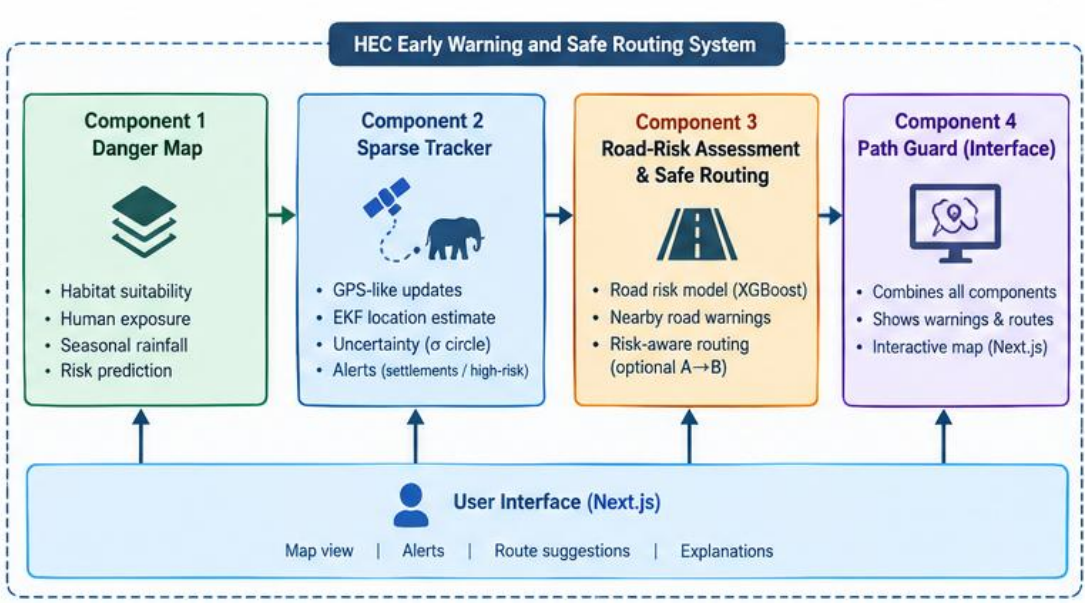


Figure 1- Position of Component 3 within the connected HEC framework

4.3 Design alternatives and decisions

The proposal describes the evolution from an earlier design based mainly on Dijkstra routing with grid-cell risk to the updated design using a separate XGBoost road-section model and a local-roads service. The updated design was selected to represent risk on individual road corridors and to support advice that names roads near the estimated elephant. The optional risk-weighted Dijkstra method is retained for route comparison.

The software-only approach was selected instead of field sensors or camera traps because the proposal prioritises a practical student prototype using public data and controlled simulation. Proxy labels are used because official conflict incident records are not used. This choice enables development within the stated scope but limits the strength of real-world validation claims.

4.4 Detailed design

4.4.1 Road network and road sections

OpenStreetMap roads are prepared as a connected road network and divided into road sections for machine learning. The final implementation description should

document how road sections are defined, which OSM tags are retained, how geometry is cleaned, how missing attributes are handled, and how section lengths are calculated.

4.4.2 Road-section feature design

The proposal identifies land and risk values from Component 1, road/highway type, hour of day, and distance from or overlap with the Component 2 uncertainty circle as inputs to the XGBoost road-section model. The final thesis should include the exact feature names, types, units, preprocessing, and any feature-selection decisions from the actual pipeline.

4.4.3 Proxy-label design

Because official incident records are not used, the training labels are proxy labels derived from high-risk corridors in Component 1 together with indicators of farmland and settlement edges. These labels represent the project's designed proxy for conflict exposure. They must not be described as verified encounters or confirmed incident data.

4.4.4 Local-roads warning logic

The primary demonstration API colours nearby roads red or green. According to the proposal, red is used for roads inside or close to the uncertainty circle, or for roads with extremely high land risk. Roads farther away are green unless land risk is extreme. The response includes structured advice in clear English, including a headline, road names to avoid, and nearby places. The precise distance thresholds and land-risk cut-offs should be documented from the implementation.

4.4.5 Optional risk-weighted routing

The proposal defines the cost of a road section for the optional A-to-B method as:

$$C(e) = \text{length}(e) \times (1 + \lambda \times \text{segment_risk}(e))$$

Here, length(e) is the physical length of a road section, segment_risk(e) is the model's proxy risk score, and λ controls how strongly risk is penalised relative to distance. The cost is used by Dijkstra's algorithm to select a minimum-cost route. The actual λ values and risk-score scaling must be taken from the implementation and experiments.

4.5 Standards and professional practice

The implementation should preserve source attribution and applicable usage conditions for public datasets, record preprocessing and model decisions, document API contracts, and distinguish predicted proxy risk from observed incidents. The prototype should communicate its limitations and avoid statements that imply a route is guaranteed safe. The final thesis should cite the relevant dataset documentation and technical references using the citation style required by the department.

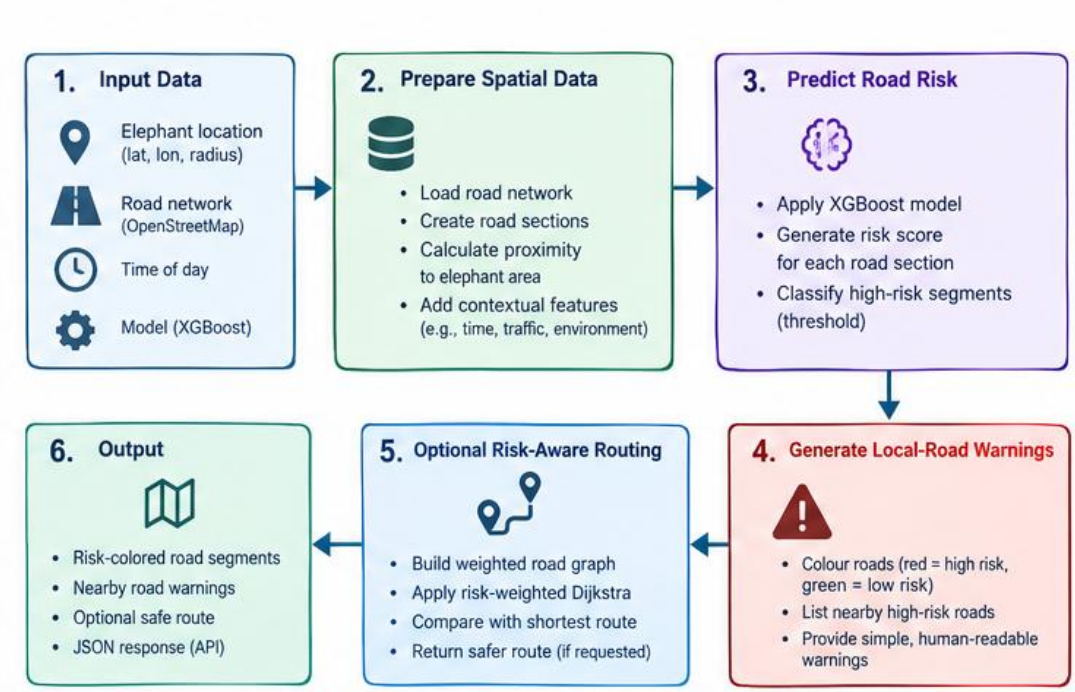


Figure 2. Component 3 road-risk assessment workflow

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The proposal identifies the overall framework as a software-only prototype. Component 3 uses OpenStreetMap road data, a separate XGBoost road-section model, a local-roads API, and an optional risk-weighted Dijkstra routing method. The overall connected application uses a Next.js map interface. The proposal also identifies a Node/Express API, Socket.io updates, a local JSON road network, and an ML microservice for Component 4 Path Guard; these should not be attributed to Component 3 unless the actual Component 3 implementation uses them.

The supplied proposal does not specify every library version, operating system, development environment, model hyperparameter, or deployment configuration for Component 3. These details should be added from the project files and environment records rather than guessed.

Technology or artefact	**Role described in proposal**	**Details to record before submission**
OpenStreetMap	Public road network and vector map data.	Extract date, coverage, download method, attribution.
XGBoost	Predicts proxy risk for individual road sections.	Package version, features, labels, parameters, model validation.
Local-roads API	Highlights nearby roads and returns structured advice.	Framework, endpoint, request/response schema, error handling.
Dijkstra routing	Optional risk-weighted A-to-B route comparison.	Graph construction, edge-cost implementation, λ values.
Next.js interface	Main interface connecting Components 1–4.	Component 3 integration contract and screenshots.
Component 1/2 services	Provide risk map and estimated location/uncertainty.	Input formats, coordinate system, update timing.

Table-05

5.2 Component implementation

5.2.1 Data preparation

The road extract is converted into a connected network and represented as road sections for modelling. Each road section should be associated with the required road attributes and spatially aligned values from the upstream risk surface. The final implementation section must describe the actual projection, geometry handling, missing-data rules, feature extraction, and road-name handling. The proposal identifies a 500 m risk grid for Component 1 and EPSG:5234 for that grid; the coordinate system actually used in Component 3 must be confirmed from its code.

5.2.2 Model training and prediction

The separate XGBoost model predicts segment_risk for road sections. Training labels are proxies built from high-risk corridors and farmland/settlement-edge indicators, not official incident records. The final thesis should report the actual model training procedure, training/validation split, random seed if applicable, model parameters, selected features, and any model evaluation metrics available from logs. These are not fully specified in the proposal.

5.2.3 Local-roads API

The local-roads service accepts an estimated elephant location and uses the surrounding uncertainty area and land-risk context to identify nearby road sections for display. The output supports red/green road colouring and structured advice in plain English. The final report should include a genuine request example, response example, endpoint specification, and screenshots captured from the running prototype.

5.2.4 Optional routing

The optional A-to-B route method uses Dijkstra with a risk-weighted cost of length × (1 + λ × segment_risk). A fair comparison should calculate the shortest and risk-weighted routes on the same road network and origin–destination pair. The selected

route, physical length, cumulative cost, and exposure to high-risk sections should be recorded for each scenario.

5.3 Integration with the other components

Component 3 receives risk context from Component 1 and estimated elephant location/uncertainty from Component 2. It produces nearby-road colours and advice. Component 4 Path Guard copies or synchronises the Component 3 road network, adds tracker-based threat circles, and compares shortest and safer A-to-B routes. All four components are connected through the main Next.js interface.

Integration tests should verify that the location and uncertainty values are passed in the expected coordinate system, road-risk responses are usable by the interface, and errors or unavailable services are communicated. The proposal describes the architecture but does not provide a full integration test report, so actual test outcomes must be added.

5.4 Verification and software testing

Test ID	Test purpose	Expected result	Actual result
T1	Load road data for the selected demonstration area.	Road network is available and has the expected geometry/attributes.	Road network data was loaded and checked for valid geometry and required attributes.
T2	Request local-road evaluation for a valid simulated elephant location.	Service returns road risk/colour information without an unhandled error.	The system returned local-road risk information for a valid simulated elephant location.
T3	Change the estimated location or uncertainty circle.	Nearby-road warnings respond according to the documented rules.	Road warnings were updated based on the estimated elephant location and uncertainty area.

T4	Request advice for a location with named nearby roads.	Response includes the expected structured advice and available road names.	The system generated structured road-risk advice and displayed available road names.
T5	Run shortest route for a fixed origin–destination pair.	A valid route and physical distance are returned.	The system generated a shortest route between the selected origin and destination and calculated its distance.
T6	Run risk-weighted route for the same pair.	A route is returned using the documented edge-cost formula.	The system generated a risk-aware route using the configured edge-cost formula.
T7	Compare shortest and risk-weighted routes.	Distance, cost, and high-risk exposure can be calculated consistently.	The system compared both routes using distance, adjusted cost, and high-risk exposure metrics.
T8	Test invalid or missing inputs and unavailable files/services.	Errors are handled and communicated appropriately.	The system handled invalid inputs and unavailable resources by returning appropriate error messages.
T9	Connect Component 3 outputs to the main interface.	Road warnings and route outputs are displayed correctly.	Component 3 outputs were checked for correct display in the main interface.

Table-06

5.5 Deployment and operation

The proposal describes a software demonstration in which the components work together through a Next.js map interface. The system uses recorded simulation

playback and public data. The final thesis should document the actual run instructions, service addresses, configuration variables, data/model file locations, startup sequence, and any limitations encountered during demonstration. It should not claim a production deployment unless one has actually occurred.

6 RESULTS AND VALIDATION

6.1 Research results

The supplied proposal describes the current design of Component 3 but does not report a completed set of numerical evaluation results for the road-section model, local-roads warnings, or optional routing comparison. This section therefore defines the results that should be reported and leaves measured values to be added after the experiments are executed.

6.1.1 Experimental setup

The fixed demonstration district is Anuradhapura District. The road network is sourced from OpenStreetMap. The road-section model is a separate XGBoost model, and its proxy labels are generated from high-risk corridors and indicators such as farmland and settlement edges. The local-roads component uses the estimated elephant location and uncertainty information from Component 2. The final report must record the exact data extracts, preprocessing, model configuration, risk threshold, uncertainty handling, software versions, λ values, origin–destination pairs, and number of completed scenarios.

Experimental detail	**Information supported by proposal**	**Information to complete from implementation**
Study area	Anuradhapura District.	Exact geographic boundary/coverage used for road extract.
Road data source	OpenStreetMap (Overpass / Geofabrik).	Actual source, retrieval date, extract version, and download method.
HEC risk context	Component 1 seasonal risk map.	Actual layer/files and processing used in Component 3.
Tracker uncertainty	Component 2 EKF uncertainty circle.	Exact format, radius calculation, and feature definition.

Model	Separate XGBoost road-section model.	Model version, features, parameters, training/validation procedure.
Routing cost	length × (1 + λ × segment_risk).	Actual λ values and risk-score scaling.
High-risk threshold	Proposal refers to extremely high land risk/high-risk corridors.	Exact threshold and how it is applied in evaluation.
Number of scenarios	Not stated.	Actual completed scenario count.

Table-07

6.1.2 Road-warning and road-section risk results

The evaluation should record the predicted segment_risk values for tested road sections and the corresponding red/green classifications returned by the local-roads service. Test cases should vary the simulated elephant location and its uncertainty area to check whether nearby-road warnings respond to those inputs. The report should also record whether the service returns a headline, road names to avoid, and nearby places where the required source data are available.

Results to insert after testing: [Insert number of road sections evaluated, model evaluation metrics if available, test scenarios, warning outputs, and representative examples. State clearly that proxy labels are not confirmed HEC incidents.]

6.1.3 Comparison of route distance

For each tested origin–destination pair, compare the conventional shortest route with the optional risk-weighted route. The additional distance is:

$$\Delta D = D(\text{risk-aware}) - D(\text{shortest})$$

The percentage increase in distance is:

$$\Delta D\% = [(D(\text{risk-aware}) - D(\text{shortest})) / D(\text{shortest})] \times 100$$

Results to insert after testing: [Insert origin–destination pairs, shortest-route distances, risk-aware route distances, additional distances, percentage increases, and observations.]

6.1.4 Comparison of high-risk route exposure

For both routes in each scenario, calculate the proportion of route distance that lies within sections classified as high risk:

$$H(P) = [D_{high\text{-}risk}(P) / D_{total}(P)] \times 100$$

The same risk surface and high-risk threshold must be used for both routes in a scenario. A lower value indicates lower exposure to sections classified as high risk by the model, but it does not establish real-world safety.

Results to insert after testing: [Insert high-risk route share for each route, absolute difference in percentage points, and observations for every tested scenario.]

6.1.5 Sensitivity to the risk-weight parameter

The proposal defines λ as the parameter controlling how strongly the risk-weighted routing cost penalises segment risk relative to distance. The sensitivity experiment should test predefined λ values while holding the road network, origin–destination pair, risk scores, and other relevant settings constant. For each value, record the selected route, distance, cumulative route cost, and high-risk route share.

Results to insert after testing: [Insert the actual λ values tested, route changes, route distances, risk exposure, and cost values. Do not report a parameter value as tested unless it was executed.]

6.2 Product and technical results

The proposal specifies a local-roads API and a separate XGBoost road-section model. The product evidence for this section should include actual endpoint tests, example request and response payloads, model-file information, road-warning screenshots, and route-comparison outputs if the optional routing method is implemented and

tested. The proposal alone does not provide these artefacts or enough evidence to claim a particular response time, accuracy score, or service-availability rate.

Insert after testing: [API endpoint evidence, screenshots, model output examples, software versions, observed errors and resolutions, and any measured performance results.]

6.3 User or stakeholder validation

The proposal describes advice written in clear English for a demonstration but does not report a completed user study or stakeholder evaluation. If a user evaluation is conducted, document the participants, procedure, consent process, questions or tasks, analysis method, and results. If no user study is conducted, state that stakeholder validation was outside the completed evaluation rather than implying that users confirmed the usefulness of the advice.

6.4 Integration results

The overall system is designed to connect Components 1, 2, 3, and 4 in one Next.js interface. Integration results should document whether Component 3 successfully receives the required risk and tracker inputs and returns road colours/advice that the interface can display. Route outputs from the optional Component 3 method should be distinguished from the separate Path Guard route comparison in Component 4.

Insert after integration tests: [List completed integration scenarios, expected and actual results, issues discovered, and screenshots or logs.]

6.5 Summary of findings

At this stage, the supplied proposal supports a description of the Component 3 design and its planned evaluation but does not establish quantitative performance for the road model or routing method. The final summary must be written after the experiments are completed. It should state which objectives were achieved, which metrics improved or did not improve, and what the evidence can and cannot support.

7 DISCUSSION

7.1 Interpretation of findings

The final interpretation must be based on measured outputs from the experiments in Chapter 6. If the local-roads service consistently marks roads according to the specified proximity and land-risk rules, this would support the technical correctness of those rules in the tested scenarios. It would not, by itself, prove that the warnings predict real elephant encounters.

If the risk-weighted route has a lower high-risk route share than the shortest route, the result would indicate reduced exposure to model-classified high-risk sections for those scenarios. If it also increases route distance, the extra distance should be reported as a trade-off. If the risk-weighted route does not reduce exposure, that outcome should be reported and investigated rather than hidden.

7.2 Comparison with literature baselines and existing solutions

The proposal motivates Component 3 by contrasting road-level risk modelling with static hotspot maps and distance-only routing. The most direct baseline is the conventional shortest route on the same road network. The risk-weighted route is then evaluated for its distance cost and high-risk route share. Any comparison with published systems or research studies requires complete source references and comparable evaluation conditions; the supplied proposal does not provide enough details to claim that this component outperforms a named external system.

7.3 Research contribution

The intended research contribution is a repeatable, software-only process that connects a public road network with a modelled HEC risk surface and an estimated elephant location to produce road-section risk scores and understandable road warnings. The optional risk-weighted route comparison extends the component from highlighting nearby roads to exploring the trade-off between route distance and model-estimated exposure.

7.4 Product contribution

The product contribution is the Component 3 road-corridor service: a road-section risk model, a local-roads API, red/green road highlighting, and structured English advice that identifies roads to avoid and nearby places where data permit. The optional risk-weighted Dijkstra method is a supporting capability. The Path Guard service and main interface belong to Component 4 and should be credited and described separately.

7.5 Limitations and threats to validity

- Proxy labels are generated from spatial risk and land-use indicators rather than confirmed HEC incidents, so model scores do not establish actual incident probabilities.
- The tracker input is based on simulated movement and an EKF uncertainty circle; it is not real-time tracking of a particular wild elephant in Sri Lanka.
- Results depend on the coverage, completeness, date, and accuracy of the public road and geographic datasets.
- Road-warning thresholds and the definition of high-risk sections can affect the resulting colours and route-exposure metrics.
- Route comparison results depend on the chosen origin–destination pairs, risk scores, λ setting, graph preparation, and edge-cost implementation.
- The proposal does not provide completed Component 3 evaluation metrics, user-validation findings, or production reliability evidence.
- The prototype is a research demonstration and is not presented as a replacement for official wildlife-warning systems.

7.6 Implications

The component illustrates how public geographic information and simulated movement can be connected to road-level decision-support outputs without installing field equipment. The approach may be useful for research demonstrations and for investigating how route-distance trade-offs change under different proxy-risk

settings. Its outputs should remain clearly labelled as model estimates and should not be used as guaranteed safety instructions without further validation.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This thesis focuses on Component 3 of a software-only HEC early-warning and safe-routing framework for Anuradhapura District. The component uses public OpenStreetMap roads and a separate XGBoost model to estimate proxy risk for road sections. The model draws on land/risk information from Component 1, road attributes, time of day, and distance from or overlap with the uncertainty circle produced by Component 2. The local-roads service highlights nearby road sections and provides understandable advice. An optional risk-weighted Dijkstra method compares routes using a cost that combines road length and segment risk.

The design is intended to bridge the gap between broad geographic risk information and road-specific warnings. However, the proposal does not provide completed quantitative results for Component 3. Final conclusions about model performance, reduced high-risk route exposure, distance overhead, or technical reliability must therefore be based on experiments and documented evidence added to the final thesis.

8.2 Achievement of objectives

Objective	**Evidence currently available in proposal**
Prepare an OpenStreetMap road network.	OSM road network is part of the stated design.
Use XGBoost to estimate road-section proxy risk.	A separate XGBoost road-section model is described.
Use Component 1 risk and Component 2 uncertainty information.	Inputs are specified in the proposal.
Provide local road colours and English advice.	Local-roads service and output behaviour are described.

Compare shortest and risk-weighted routes.	Optional Dijkstra cost formula is specified.
Evaluate the component without field equipment or restricted collar data.	Software-only public-data and simulation scope is explicit.

Table-08

8.3 Individual research and product contributions

The individual research contribution is the design and evaluation of a road-corridor risk approach that combines public road geometry, spatial risk context, and simulated location uncertainty. The product contribution is the local-roads risk service and optional risk-weighted route comparison. The final version should be updated to identify the exact code, model, data-processing, testing, and integration work completed by the author.

8.4 Future directions

- Complete and document Component 3 model evaluation using an appropriate validation procedure for the proxy labels.
- Test multiple simulated elephant locations, times of day, uncertainty settings, and origin–destination pairs.
- Compare shortest and risk-weighted routes using route distance, high-risk route share, and cumulative weighted cost.
- Investigate sensitivity to the risk penalty λ and the thresholds used to classify high-risk road sections.
- Improve the traceability of advice by recording the road attributes and risk conditions that led to each warning.
- Conduct a carefully designed usability evaluation of whether the English advice is understandable to intended users.
- Explore whether additional verified data could support stronger validation, while respecting data access, privacy, and wildlife-conservation constraints.

- Keep the system clearly labelled as a research prototype until external and real-world validation are available.

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GLOSSARY

HEC — Human–Elephant Conflict.

EKF — Extended Kalman Filter.

OSM — OpenStreetMap.

CRS — Coordinate Reference System.

XGBoost — Extreme Gradient Boosting (machine-learning algorithm used in Component 1).

AUC — Area Under the Receiver Operating Characteristic Curve (model performance metric).

Dijkstra — Shortest-path algorithm for graphs with non-negative edge weights.

GeoJSON — Open standard format for encoding geographic data structures.

API — Application Programming Interface.

APPENDICES

Appendix A Research instruments or extended results

Field	Record for each experiment
Scenario ID	SR-01
Study area and road extract	Anuradhapura District, OpenStreetMap-derived road network; the exact extract date and version still need recording
Simulated elephant location	Longitude: 80.40° E; Latitude: 8.35° N (WGS84)
Uncertainty representation	API request uses 800 m; the API default is 500 m.
Hour of day	18:00
Risk-model version	road_risk_model.joblib, the XGBoost road-risk classifier
λ value	Record the value configured in src/config.py and used by run_route.py for the risk-weighted routing experiment
Shortest route distance	To be recorded from the actual shortest-route output (metres or kilometres).
Risk-weighted route distance	To be recorded from the actual risk-weighted route output (metres or kilometres).
High-risk route share for shortest route	To be calculated from the measured distance within high-risk road segments divided by the total shortest-route distance, multiplied by 100.
High-risk route share for risk-weighted route	To be calculated from the measured distance within high-risk road segments divided by the total risk-weighted route distance, multiplied by 100.
Route cost	Record the actual calculated cost of each route and specify the cost units or interpretation used by the implementation.

Table-9

Appendix B Academic integrity and AI use disclosure

Complete this appendix in accordance with the module's academic integrity policy. State the tools used, their purpose, representative prompts or workflows where required, the verification performed, and which thesis sections or artefacts were affected. The student remains responsible for checking all statements against the proposal, code, data, experimental outputs, and source references.